

CPEN 416 / EECE 5710

Lecture 21: Stabilizer codes

Tuesday 25 November 2025

Announcements

- Quiz 9 today
- TA4 due Friday 23:59
- One more content lecture (Thursday), then presentations

↓
group 03-05 - can
anyone switch w/group 01?

Last time

We made a code to fix phase flip errors by making them look like bit flip errors in a different basis.

$$\text{Bit flip:} \quad \alpha |000\rangle + \beta |111\rangle$$

$$\text{Phase flip:} \quad \alpha |+++ \rangle + \beta |-- - \rangle$$

Learning outcomes

Today:

- ➊ Outline the conditions under which errors can be corrected
- ➋ Define the stabilizers of a quantum error correcting code
- ➌ Express the bit flip, phase flip, Shor code, and 5-qubit code in the stabilizer formalism

Correcting arbitrary errors

The 9-qubit code can correct *arbitrary* errors on a *single qubit*.

Represent possible errors as $\{E_i\}$, full error channel as

$$\mathcal{E}(|\psi\rangle) = \sum_i E_i |\psi\rangle E_i^\dagger$$

We can correct $\{E_i\} = \{I, X_j, Z_j, X_j Z_j\}_j$. If we can write an arbitrary error E_k on qubit j as

$$E_{k,j} = e_{k_0} I + e_{k_1} X_j + e_{k_2} Z_j + e_{k_3} X_j Z_j$$

then we can correct it too.

Conditions for quantum error correction

How do we know if a particular error is correctable by a QECC?

Formal definition of a quantum error correcting code is a *subspace*, C , called the **codespace**.

$$P(\alpha|010\rangle + \beta|101\rangle) = 0$$

Example: bit flip code.

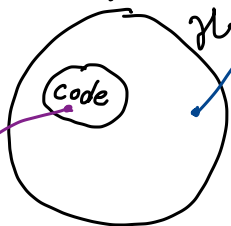
■ Codewords: $|000\rangle$, $|111\rangle$

■ Codespace: contents have the form $\alpha|000\rangle + \beta|111\rangle$

Define a **projector** onto the codespace,

$$P = |000\rangle\langle 000| + |111\rangle\langle 111|$$

$$P(\alpha|000\rangle + \beta|111\rangle) = \alpha|000\rangle + \beta|111\rangle$$

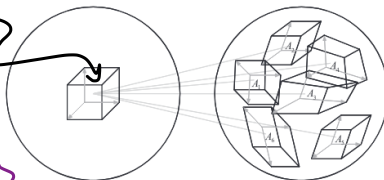


Conditions for quantum error correction

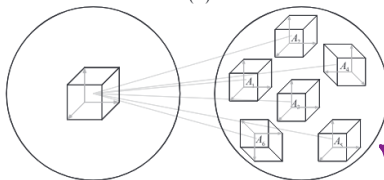
The errors this code corrects are all mapped to different **orthogonal** subspaces of the full Hilbert space.

No error
 $\alpha |000\rangle + \beta |111\rangle$

$\alpha |100\rangle + \beta |011\rangle$



(A)



(B)

Errors :

$$\alpha |100\rangle + \beta |011\rangle$$

$$\alpha |010\rangle + \beta |101\rangle$$

$$\alpha |001\rangle + \beta |110\rangle$$



11

Conditions for quantum error correction

Theorem 10.1: (Quantum error-correction conditions) Let C be a quantum code, and let P be the projector onto C . Suppose $\mathcal{E}$ is a quantum operation with operation elements $\{E_i\}$. A necessary and sufficient condition for the existence of an error-correction operation $\mathcal{R}$ correcting $\mathcal{E}$ on C is that

$$\underbrace{P E_i^\dagger}_{\text{for some Hermitian matrix } \alpha} \underbrace{E_j P}_{\text{of complex numbers}} = \alpha_{ij} P \quad (10.16)$$

If such an $\mathcal{R}$ exists, $\{E_i\}$ is called a *correctable set of errors*.

Correctable errors: bit flip code

$$P = |000\rangle\langle 000| + |111\rangle\langle 111|$$

Consider $\{I, E_1 = XII, E_2 = IXI, E_3 = IIX\}$ $I \otimes I \otimes X$

$$\begin{aligned} \underbrace{P E_2^\dagger E_3 P} &= (|000\rangle\langle 000| + |111\rangle\langle 111|) I X I \cdot I I X (|000\rangle\langle 000| + |111\rangle\langle 111|) \\ &= (|000\rangle\langle 010| + |111\rangle\langle 101|) (|001\rangle\langle 000| + |110\rangle\langle 111|) \\ &= 0 \cdot P \end{aligned}$$

$$P E_j^\dagger E_i P = \delta_{ij} \cdot P$$

$$P E_2^\dagger E_2 P = P$$

Try including a 2-qubit error: $E_4 = XXII$

$$\begin{aligned} P (IIX) (XXI) P &= P (XXX) P \\ &= P \cdot (|111\rangle\langle 000| + |000\rangle\langle 111|) \\ &= |000\rangle\langle 111| + |111\rangle\langle 000| \neq \alpha \cdot P \end{aligned}$$

Example: amplitude damping

Not immediately clear how to express this in terms of no errors, bit, phase or bit/phase flip errors. Can we still correct it?

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

Straightforward to re-express K_1 :

$$K_1 = \frac{\sqrt{p}}{2} (X + iY)$$

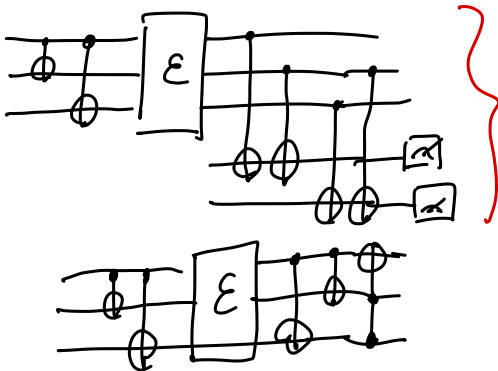
For K_0 ,

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & (1-p)^{1/2} \end{pmatrix} \approx \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{p}{2} + \overset{\text{big } 0}{O(p^2)} \end{pmatrix}$$

$$= \left(1 - \frac{p}{4}\right) I + \frac{p}{4} Z + \dots$$

Bit flip code: recovery revisited

Last week / last class, we considered two recovery circuits for the bit flip code.



Stabilizers

We can consider a more general invariant than parity: eigenvalues w.r.t. special subsets of the Pauli group.

The n -qubit Pauli group is

$$\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n} \times \{1, -1, i, -i\}$$

$$Y|0\rangle = i|1\rangle$$

Example: Which two-qubit Paulis is this Bell state a +1 eigenstate of?

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$\begin{array}{c} II \\ IX \\ \vdots \\ ZY \\ ZZ \end{array}$$

$$P|\psi\rangle = |\psi\rangle \Rightarrow P = \{II, ZZ, XX\}$$

Bit flip code: stabilizers

Consider our logical states:

$$|0\rangle_L = |000\rangle$$

$$|1\rangle_L = |111\rangle$$

$$P|000\rangle = |000\rangle$$

$$P|111\rangle = |111\rangle$$

Which three-qubit Paulis are these states +1 eigenstates of?

$$III|000\rangle = +1 \cdot |000\rangle$$

$$\{ZZI, ZIZ, IZZ\}$$

Stabilizers

Let S be a subgroup of $\mathcal{P}_n$.

Let V_S be a set of states that are $+1$ eigenstates for all $P \in S$.

Then, S is the **stabilizer** of V_S , and V_S is stabilized by S .

Facts about S :

- $-I$ is never in S
- all items of S commute
- choosing S uniquely defines the fixed subspace, V_S

Next time

Next class (last class):

- More on stabilizer codes; fault-tolerant quantum computing

Action items:

- ① TA4 due Friday at 23:59
- ② Work on project

Recommended reading:

- From this class: Codebook EC; N&C 10.1-10.5
- For next class: Codebook EC; N&C 10.6